Training set (N=3000) Validation set (N=1000) Testing set (N=1000) Zero error line (y=0) ±1σ test residual band

(a) CatBoost

Test μ_{res} = -0.14 kN·m
Test σ_{res} = 2.24 kN·m
Test RMSE = 2.25 kN·m

(b) MLP

Test μ_{res} = 0.03 kN·m
Test σ_{res} = 2.72 kN·m
Test RMSE = 2.72 kN·m

(c) XGBoost

Test μ_{res} = -0.12 kN·m
Test σ_{res} = 3.02 kN·m
Test RMSE = 3.03 kN·m

(d) LightGBM

Test μ_{res} = -0.35 kN·m
Test σ_{res} = 3.56 kN·m
Test RMSE = 3.58 kN·m

(e) SVR

Test μ_{res} = 0.07 kN·m
Test σ_{res} = 3.64 kN·m
Test RMSE = 3.64 kN·m

(f) RF

Test μ_{res} = -0.82 kN·m
Test σ_{res} = 6.57 kN·m
Test RMSE = 6.62 kN·m

(g) DT

Test μ_{res} = -0.09 kN·m
Test σ_{res} = 10.59 kN·m
Test RMSE = 10.59 kN·m

(h) Ridge

Test μ_{res} = 0.41 kN·m
Test σ_{res} = 11.97 kN·m
Test RMSE = 11.98 kN·m

Residual (kN·m)

Predicted T_u (kN·m)